Data-driven Research in Structural Bioinformatics: from Proteins to Biomolecular Interactions

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2026-10-07



Life Sciences Center



Institute of Biotechnology

We study molecules of life computationally

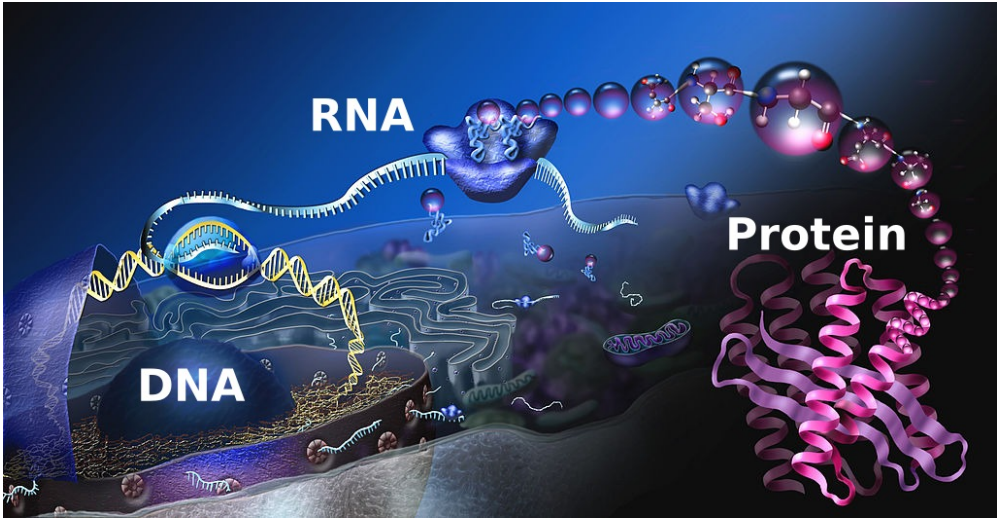
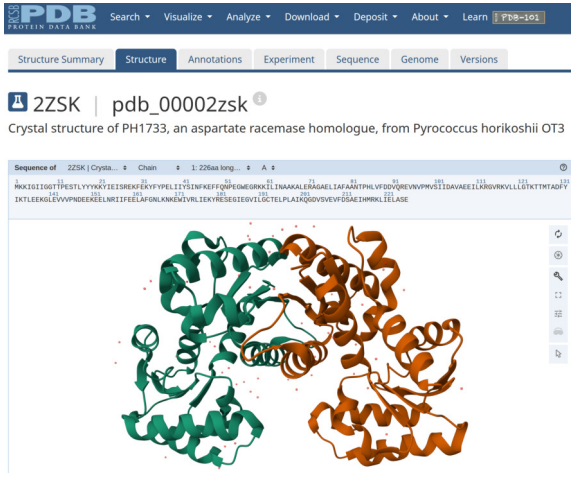


Image source: jonlieffmd.com/blog/new-studies-reveal-higher-levels-of-genetic-complexity

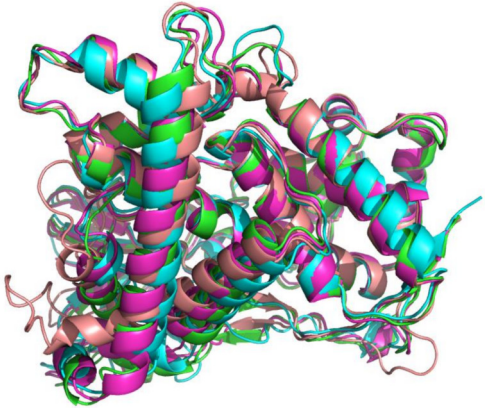
Protein Data Bank (PDB) is FAIR

- **F - Findable**: stable PDB identifiers, searchable metadata, links to publications and external databases.
- **A - Accessible**: openly available, downloadable in bulk, accessible programmatically.
- **I - Interoperable**: standardized formats, sequence–structure mapping, cross-references to UniProt.
- **R - Reusable**: provenance and versioning are maintained, data can be systematically reprocessed into new resources and datasets.



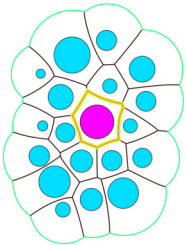
Multiple sequence alignment (MSA) as a natural interoperability layer

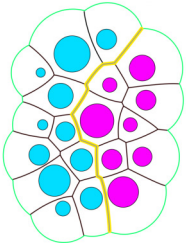




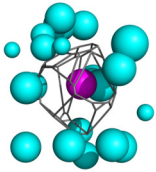
We use Voronoi tessellation to get from raw coordinates to interactions

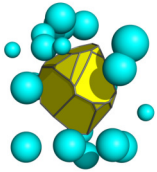
2D examples

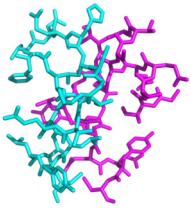


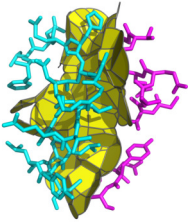


3D examples

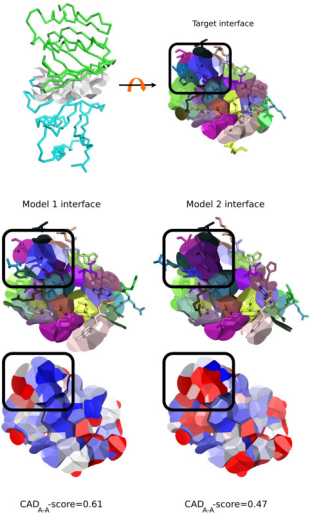






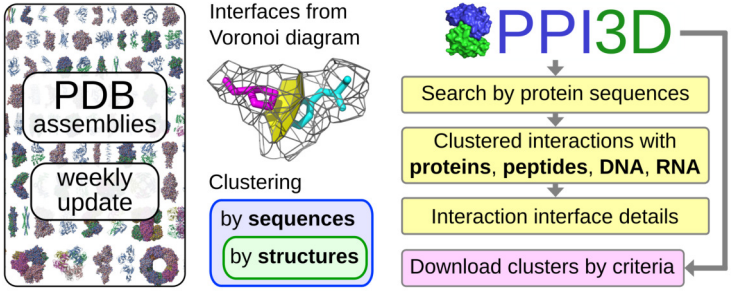


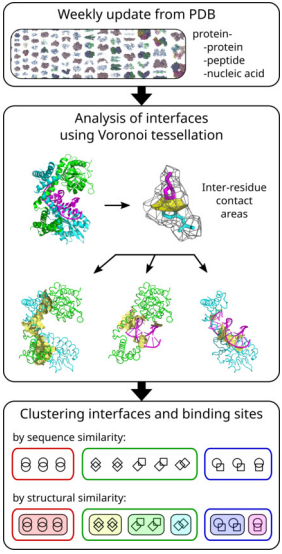
Contact Area Difference score:



Protein-protein Interactions in 3D (PPI3D)

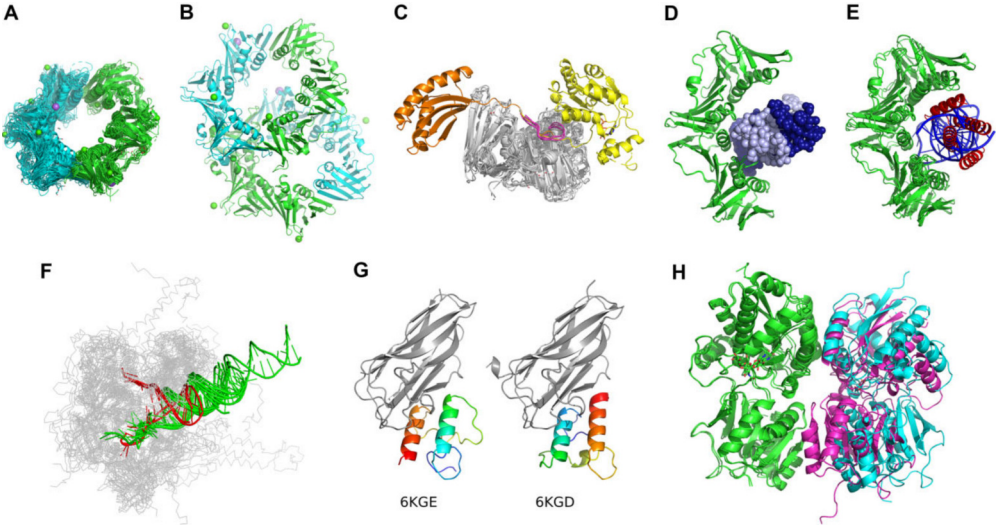
https://bioinformatics.lt/ppi3d/





Dapkūnas J, Timinskas A, Olechnovič K, Tomkuvienė M, Venclovas Č. *PPI3D: a web server for searching, analyzing and modeling protein-protein, protein-peptide and protein-nucleic acid interactions*. Nucleic Acids Res, 2024

Examples of PPI3D analyses

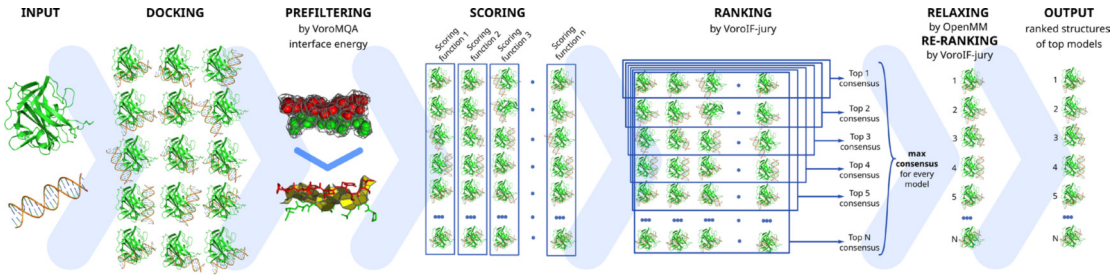


PPI3D is FAIR

- **F - Findable**: search interactions by one protein sequence, two sequences, or a PDB entry, results are organized into non-redundant interaction/interface clusters.
- **A - Accessible**: openly available, and users can download custom sets of interaction interfaces for local analysis.
- **I - Interoperable**: uses a consistent interface definition, sequence clustering and MSAs to map corresponding residues across homologous proteins, so interfaces from different structures can be compared in the same framework.
- **R - Reusable**: the database is updated weekly from the PDB, reduces redundancy while preserving alternative interaction modes, provides downloadable datasets suitable for large-scale analysis and ML.

FAIR structural data can be transformed into FAIR interaction data.

FTDMP framework for protein-protein/RNA/DNA docking and scoring



Olechnovič K, Banciul R, Dapkūnas J, Venclovas Č. *FTDMP: A Framework for Protein-Protein, Protein-DNA, and Protein-RNA Docking and Scoring*. Proteins, 2024

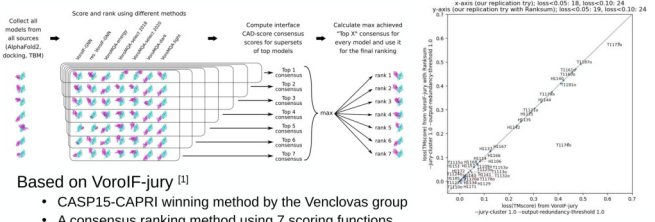
FTDMP was successfully used in CASP and CAPRI challenges

https://predictioncenter.org/casp16/zscores_multimer.cgi

GR name	GR code	Targets Count	SUM Zscore (>-2.0)	Rank SUM Zscore (>-2.0)
KiharaLab	294	37	15.3597	1
Yang-Multimer	456	37	14.5255	2
Yang-Server	052	37	13.5255	3
MULTICOM_human	345	37	13.0386	4
Yang	022	36	12.9135	5
MIEnsembles-Server	110	37	12.0918	6
Zheng	462	37	10.9631	7
MULTICOM_GATE	425	37	10.3399	8
MULTICOM_LLM	319	37	10.2300	9
AF3-server	304	37	9.9460	10
kozakovvajda	274	34	9.4231	11
Zheng-Multimer	147	37	9.4126	12
MULTICOM_AI	331	37	9.1939	13
MULTICOM	051	37	9.1741	14
Huang-HUST	091	37	8.1268	15
elofsson	241	37	8.0457	16
NKRNA-s	028	37	7.6095	17
GHZ-ISM	075	37	5.3304	18
CSSB-Human	419	37	5.0413	19
plmfold	287	36	4.2086	20

CASP16 presentation slide from KiharaLab (Purdue, USA):

Scoring: VoroIF-jury + LZerD Ranksum



Based on VoroIF-jury [1]

- CASP15-CAPRI winning method by the Venclovas group
- A consensus ranking method using 7 scoring functions

Added three scoring functions (components of LZerD RankSum score)

- GOAP[2], DFIRE[3], and ITScorePro[4]

PURDUE UNIVERSITY

[1] Olechnovič et al., Proteins, 2023
[2] Zhou et al. Biophys. J., (2011)
[3] Zhou et al. Protein Sci., (2002)
[4] Huang et al. Proteins (2002)

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Published benchmarks ≠ ready-to-reuse benchmarks

Previously published docking benchmarks

- **Protein–protein** — 252 cases
  Vreven et al. (2015)
  Guest et al. (2021)
- **Protein–DNA** — 47 cases
  van Dijk & Bonvin (2008)
- **Protein–RNA** — 42 cases
  Pérez-Cano et al. (2012)
  Huang & Zou (2013)
  Nithin et al. (2017)

Before testing FTDMP, we had to . . .

- re-download structures from the **PDB**
- select the correct chains
- standardize residue numbering
- check alternative binding modes with PPI3D
- correct inconsistencies and errors
- normalize modified nucleotides

Accessible data are not necessarily interoperable or reusable.

VoroMarmotte: hidden information in the PDB

Ensembles of PDB structures of related sequences

↓

MSA-based residue mapping

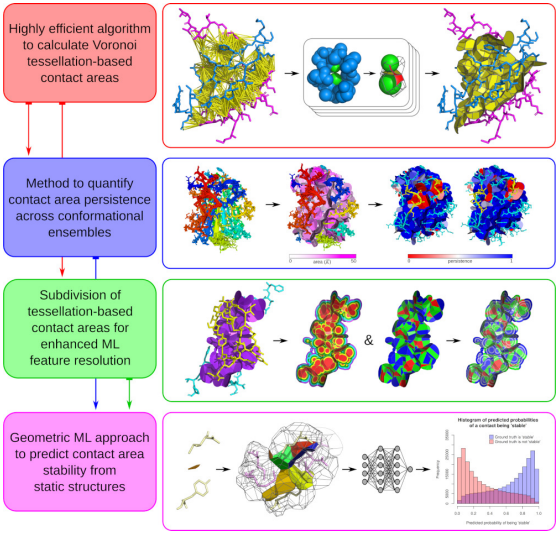
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Tessellation-derived contact area persistence

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Machine learning

Reusable data can reveal information that was never explicitly deposited.



TemStaPro: learning protein thermostability at scale

Pudžiuvelytė I, Olechnovič K, Godliauskaite E, Sermokas K, Urbaitis T, Gasiunas G, Kazlauskas D. *TemStaPro: protein thermostability prediction using sequence representations from protein language models.* Bioinformatics, 2024.

Idea: use organism growth temperature as a proxy for protein stability.

Growth temperature is not the exact protein melting temperature, but provides a useful lower-bound type of label. TemStaPro therefore learns multiple temperature-threshold classifiers.

How to obtain enough labelled sequences?

Organism metadata (21,498 organisms with annotated growth temperatures and taxonomy IDs)

$+$

Protein sequence data (UniParc proteomes linked by taxonomy)

$\downarrow$

$>$1 million labelled protein sequences

Two interoperable FAIR resources $\rightarrow$ a new ML dataset

SKEMPI: well curated, but maybe too small for machine learning

Jankauskaitė J, Jiménez-García B, Dapkūnas J, Fernández-Recio J, Moal IH. *SKEMPI 2.0: an updated benchmark of changes in protein–protein binding energy, kinetics and thermodynamics upon mutation*. Bioinformatics, 2019.

What we would really like:

protein complex + mutation $\longrightarrow$ change in binding affinity

SKEMPI 2.0

- manually curated experimental data
- structure-linked mutations
- binding affinity, kinetics, thermodynamics
- standardized units and residue numbering

But only . . .

- 7,085 mutation entries
- 237 protein-protein interactions
- 345 PDB structures
- substantial biological and mutation bias

For ML, good organization cannot compensate for scarce experimental measurements.

Acknowledgements

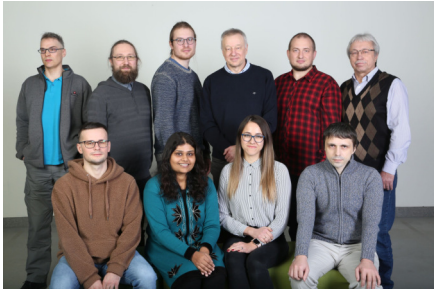
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- Dr. Darius Kazlauskas
- Dr. Mindaugas Margelevičius
- Dr. Kliment Olechnovič
- Dr. Albertas Timinskas
- Kęstutis Timinskas
- Rytis Dičiūnas
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- Rita Banciul
- Jyotika Pachauri
- Praneet Prabhanjan Bejavada
- Anastasija Kudrevceva
- Martynas Jakutis
- Shikhar Rana

Former members:

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- Ieva Pudžiuvelytė
- Lukas Valančauskas

https://bioinformatics.lt



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Funded by the European Union